MILITARY SPECIFICATION

CARTRIDGE 22MM, SUB-CALIBER PRACTICE M744, M745, M746, M747 LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembly and packing of one type of training round for the 81MM Mortar System.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-48077 - Primer, M104 Parts For And Loading, Assembling And Packing

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-810 - Environmental Test Methods

FSC: 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9299079	-	Cartridge 22MM, Sub-Caliber Practice M744, M745, M746, M747
9322198	-	Tray Assembly for Cartridge, Sub-Caliber, 22MM, Practice
9322199	-	Bag, Barrier
9322201	-	Box, Wirebound
9322202	-	Packing and Marking For Box, Wood, Wirebound, For Cartridge, Sub-Caliber 22MM, Practice

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Ejection and propelling charges and components and assemblies

3.2.1 Ejection and propelling charge. - The ejection and propelling charges shall comply with all requirements specified on drawing (dwg.) 9299080 or 9331155 as applicable and associated drawings and with all requirements specified in applicable specifications and standards.

3.2.1.1 Propelling and ejection charge funtioning

a. The propelling charge and the ejection charge for charge one (ref. part no. 9299080-1) shall produce a velocity of no more then 155 feet per second (FPS) nor less then 133 FPS to launch the 81MM sabot assembly at a quandrant elevation (QE) of 800 mils.

b. The propelling charge and the ejection charge for charge two (ref. part no 9299080-2) shall produce a velocity of no more than 176 FPS nor less than 142 FPS to launch the 81MM sabot assembly at a QE of 800 mils.

c. The propelling charge and the ejection charge for charge three (ref. part no. 9299080-3) shall produce a velocity

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of no more than 215 FPS nor less than 177 FPS to launch the 81MM sabot assembly at a QE of 800 mils.

d. The propelling charge and the ejection charge for charge four (ref. part no. 9299080-4) shall produce a velocity of no more than 233 FPS nor less than 214 FPS to launch the 81MM sabot assembly at a QE of 800 mils.

3.2.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing (dwg.) 9299079 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Proving ground test. The cartridges shall comply with the following requirements when tested in accordance with 4.5.2 and subjected to ambient ($70^{o}F$) temperature condition.

3.3.1 Misfires. The cartridges shall not fail to fire from the weapon.

3.3.2 Duds. The cartridges shall function on impact. The cartridges upon initiation on impact shall issue an audible report and a visible smoke spotting cloud.

3.3.3 Short rounds. No cartridge shall have a horizontal range less than 80 percent of the average horizontal range (calculated without the short round).

3.3.4 Premature burst. There shall be no premature burst in the gun bore or in flight.

3.3.5 Transportation vibration. The cartridges shall be safe and operable after being subjected to the procedures as specified in MIL-STD-810, Method 514, procedure X and XI, curve AA.

3.4 First article inspection. This specification contains technical provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.5 Data requirements. The contractor shall generate data in accordance with the requirements of the data item descriptions cited in 6.5.

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3.6 Workmanship. All parts shall be finished and assembled in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provision of 4.3.2. The first article sample shall consist of 1 box with 100 each cartridges (dwg. 9299079) and components in the quantities indicated in Table 1.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table 1 specified herein.

4.3.3 Rejection. See MIL-A-48078 and as follows for the first article ballistic testing:

a. One or more misfires occur. (Each 25 sample group)

b. One or more short rounds occur. (Each 25 sample group)

c. One or more premature bursts occur. (Each 25 sample group)

4.3.4 Referral. If five or more duds (each 25 sample group) occur the lot shall be in referred status and ARRADCOM shall be notified to the attention of DRDAR-QAR-M.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
	Cartridge 22MM, Sub-Caliber Practice M744, M745, M746, M747		1 OF 5		See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Head (Dwg. 9299110) Examination for defects	15		3.2	4.4.2.1
	Case, Cartridge Assembly (Dwg. 9299112) Examination for defects	15		3.2	4.4.2.2
	Case, Cartridge (Dwg. 9299105) Examination for defects	15		3.2	4.4.2.3
	Cup, Washer (Dwg. 9299109) Examination for defects	15		3.2	4.4.2.4
	Percussion Slide Assembly (Dwg. 9299101) Examination for defects	15		3.2	4.4.2.5
	Slide, Percussion (Dwg. 9299102) Examination for defects	15		3.2	4.4.2.6
	Case Assembly (Dwg. 9299114) Examination for defects	15		3.2	4.4.2.7

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
	Cartridge 22MM, Sub-Caliber Practice M744, M745, M746, M747		2 OF 5	See Below
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Case, Shell (Dwg. 9311005)				
	Examination for defects	15		3.2	4.4.2.8
	Disc, Closing (Dwg. 9311004)				
	Examination for defects	15		3.2	4.4.2.9
	Wing Assembly (Dwg. 9311002)				
	Examination for defects	15		3.2	4.4.2.10
	Wing (Before forming) (Dwg. 9311003)				
	Examination for defects	15		3.2	4.4.2.11
	Wing Embrittlement	15		3.2	4.5.5
	Wing (After forming) (Dwg. 9311003)				
	Examination for defects	15		3.2	4.4.2.12
	Case, Ignition (Dwg. 9299108)				
	Examination for defects	15		3.2	4.4.2.13
	Basin (Type 1, 2 & 3) (Dwg. 9299095)				
	Examination for defects	15		3.2	4.4.2.14

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
	Cartridge,22MM, Sub-Caliber Practice M744, M745, M746 M747		3 OF 5		See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Basin (Type 4)(if applicable) (Dwg. 9299100)				
	Examination for defects	15		3.2	4.4.2.15
	Nozzle, Jet (Dwg. 9299089)				
	Examination for defects	15		3.2	4.4.2.16
	Disc (Dwg. 9299090)				
	Examination for defects	15		3.2	4.4.2.17
	Foil, Damming (Dwg. 929909 3)				
	Examination for defects	15		3.2	4.4.2.18
	Jet Nozzle Assembly (Dwg. 9299088)				
	Examination for defects	15		3.2	4.4.2.19
	Jet Housing Assembly (Type 1, 2 & 3) (Dwg. 9299080)				
	Examination for defects	15		3.2	4.4.2.20
	Charge weight assessment (For each ind. type)	30		3.2	4.5.4
	Cap, Protective (Dwg. 9299097)				
	Examination for defects	15		3.2	4.4.2.21

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
	Cartridge 22MM, Sub-Caliber Practice M744, M745, M746 M747		4 OF 5	See Below
				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge, 22MM Sub-Caliber Practice M744, M745, M746, M747 (Dwg. 9299079)				
	Examination for defects	5 ea		3.2	4.4.2.22 (a)
	Proving ground test	25 ea		3.2	4.4.2.36 (a)
	Transportation vibration test	100 ea		3.3.5	4.5.2.2
	Body, Jet (Dwg. 9299084)				
	Examination for defects	15		3.2	4.4.2.24
	Charge, Ejection (Dwg. 9299098)				
	Examination for defects	4 grams		3.2	4.4.2.25
	Cap, Percussion (Dwg. 9299091)				
	Examination for defects	15		3.2	4.4.2.26
	Disc, Flat (Dwg. 9299096)				
	Examination for defects	15		3.2	4.4.2.27
	Cap, Percussion (Dwg. 9299103)				
	Examination for defects	15		3.2	4.4.2.28

NOTES: (a) The examination for defects and the proving ground test shall be performed on the same units that were subjected to the transportation vibration test.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
	Cartridge 22MM, Sub-Caliber Practice M744, M745, M746, M747		5 OF 5		See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Disc, Seal (Dwg. 9299104)				
	Examination for defects	15		3.2	4.4.2.29
	Disc (Dwg. 9299107)				
	Examination for defects	15		3.2	4.4.2.30
	Smoke Charge Assembly (Dwg. 9299111)				
	Examination for defects	15		3.2	4.4.2.31
	Charge, Smoke (Dwg. 9299113)				
	Examination for defects	14 grams			4.4.2.32
	Jet Housing Assembly (Type 4) (Dwg. 9331155)				
	Examination for defects	15		3.2	4.4.2.33
	Charge weight assessment	30		3.2	4.5.4

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Cartridges shall contain:

a. Metal parts from lots with the same interfix lot number from one manufacturer.

b. Primers from lots with the same interfix lot number from one manufacturer.

c. Smoke charge, propelling charge, and ejection charge from lots with the same interfix lot numbers from one manufacturer.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.1	Head		1 OF 1		9299110
					NEXT HIGHER ASSEMBLY 9299079
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Pitch diameter of rear threads		0.40%	3.2	Gage
103	Major diameter of rear threads		0.40%	3.2	Gage
104	Runout of rear threads with outside diameter and shoulder face forward of threads		0.40%	3.2	Gage
105	Diameter of groove		0.40%	3.2	Gage
106	Depth to shoulder forward of threads		0.40%	3.2	Gage
107	Profile of firing pin		0.40%	3.2	Gage
108	Length of protrusion of firing pin beyond aft face		0.40%	3.2	Gage
109	Depth of cavity		0.40%	3.2	Gage
110	Diameter of cavity, minimum		0.40%	3.2	Gage
111	Diameter of cavity chamber angle		0.40%	3.2	Gage
112	Radius of firing pin point		0.40%	3.2	Gage
Minor					
201	Protective finish damaged or inadequate		0.65%	3.2	Visual
202	Width of groove		0.65%	3.2	Gage
203	Chamfer angle of cavity		0.65%	3.2	Gage
204	Diameter of firing pin base		0.65%	3.2	Gage
205	Overall length of head		0.65%	3.2	Gage
206	Taper of ogive		0.65%	3.2	Gage
207	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Case, Cartridge Assembly	1 OF 1	9299112
			NEXT HIGHER ASSEMBLY 9299079

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length, max.		0.40%	3.2	Gage
102	Crimped inner diameter, max.		0.40%	3.2	Gage
103	Crimp inadequate or cracked		0.40%	3.2	Visual
104	Component parts missing or improperly assembled		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Case, Cartridge	1 OF 1	9299105
			NEXT HIGHER ASSEMBLY 9299112

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Large cavity inside diameter		0.40%	3.2	Gage
103	Pitch diameter thread		0.40%	3.2	Gage
104	Minor diameter thread		0.40%	3.2	Gage
105	Length, overall		0.40%	3.2	Gage
106	Runout of rear thread with outside diameter and aft face		0.40%	3.2	Gage
107	Length of unthreaded inside chamber		0.40%	3.2	Gage
108	True position of large cavity inside diameter with outside diameter		0.40%	3.2	Gage
109	Thickness of forward crimping surface		0.40%	3.2	Gage
Minor					
201	Radius at aft end of crimping surface improper or missing		0.65%	3.2	Gage
202	Length from forward face to aft end of radius at outside surface		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Cup, Washer	1 OF 1	9299109
			NEXT HIGHER ASSEMBLY: 9299112

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Height		0.40%	3.2	Gage
103	Wall thickness at wall base		0.40%	3.2	Gage
Minor					
201	Thickness of side wall		0.65%	3.2	Gage
202	Inside radius missing or improper		0.65%	3.2	Visual
203	Outside radius missing or improper		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9299101
4.4.2.5	Percussion Slide Assembly				NEXT HIGHER ASSEMBLY 9299112
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Primer missing or deformed		0.40%	3.2	Visual
102	Components improperly assembled		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Slide, Percussion	1 OF 1	9299102
			NEXT HIGHER ASSEMBLY 9299101

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Inside diameter of thru hole		0.40%	3.2	Gage
103	Location of ball-groove apex		0.40%	3.2	Gage
104	Angle of cutout groove		0.40%	3.2	Gage
105	Location of outside ball radius groove		0.40%	3.2	Gage
Minor					
201	Diameter of large cavity		0.65%	3.2	Gage
202	Depth of large diameter cavity		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.7	Case Assembly		1 OF 1		9299114
					NEXT HIGHER ASSEMBLY 9299111
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length, max		0.40%	3.2	Gage
102	Component parts missing or improperly assembled		0.40%	3.2	Visual
103	Length from end of shell case to wing		0.40%	3.2	Gage
104	Runout of aft diameter with forward outside diameter		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Case, Shell	1 OF 2	9311005
			NEXT HIGHER ASSEMBLY 9299114

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thread pitch diameter		0.40%	3.2	Gage
102	Thread minor diameter		0.40%	3.2	Gage
103	Outside diameter		0.40%	3.2	Gage
104	Inside diameter (wing end)		0.40%	3.2	Gage
105	Runout of rear inside diameter to outside diameter		0.40%	3.2	Gage
106	Runout of thread to outside diameter and forward face		0.40%	3.2	Gage
107	Length, overall		0.40%	3.2	Gage
108	Length from aft face to aft end of wing interface groove		0.40%	3.2	Gage
109	Length from aft face to forward end of rear cavity		0.40%	3.2	Gage
110	Depth of wing interface groove				
Minor					
201	Protective finish missing or inadequate		0.65%	3.2	Visual
202	Central inside diameter		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.8	Case, Shell		2 OF 2	9311005
				NEXT HIGHER ASSEMBLY
				9299114

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
203	Runout of central inside diameter with outside diameter		0.65%	3.2	Gage
204	Forward cavity diameter aft of the thread		0.65%	3.2	Gage
205	Length from forward face to aft end of forward cavity		0.65%	3.2	Gage
206	Depth of forward cavity		0.65%	3.2	Gage
207	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.9	Disc, Closing		1 OF 1		9311004
					NEXT HIGHER ASSEMBLY 9299114
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
Minor					
201	Protective finish missing or inadequate		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Wing Assembly	1 OF 1	9311002
			NEXT HIGHER ASSEMBLY: 9299114

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length		0.40%	3.2	Gage
102	Rear outside diameter		0.40%	3.2	Gage
103	Distance from front face to wing		0.40%	3.2	Gage
104	Forward outside diameter		0.40%	3.2	Gage
105	Runout of rear outside diameter to forward outside diameter		0.40%	3.2	Gage
106	Perpendicularity between aft face and forward outside diameter		0.40%	3.2	Gage
107	Inside diameter of cavity		0.40%	3.2	Gage
108	Distance from ends of parallel fin blades		0.40%	3.2	Gage
109	Fin blades damaged, distorted or improperly molded to body		0.40%	3.2	Visual
Minor					
201	Depth of cavity		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Wing (Before forming)	1 OF 1	9311003
			NEXT HIGHER ASSEMBLY
			9311002

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wing embrittlement	5(a)		3.2	4.5.5
102	Width of wing		0.40%	3.2	Gage
103	Radii missing or improper (4 places)		0.40%	3.2	Visual
104	Thickness		0.40%	3.2	Gage
105	Length of wing		0.40%	3.2	Gage
106	Angle cut offs improperly located or incorrect (4 places)		0.40%	3.2	Gage/Visual
Minor					
201	Location of holes		0.65%	3.2	Visual
202	Diameter of holes		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

(a) Sample per heat treat batch.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Wing (After forming)	1 OF 1	9311003
			NEXT HIGHER ASSEMBLY 9311002

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wings not normal to each other		0.40%	3.2	Gage
102	Length from kink to end of wing		0.40%	3.2	Gage
103	Angle of kink		0.40%	3.2	Gage
104	Length of kinked wing		0.40%	3.2	Gage
105	Length of unkinked wing		0.40%	3.2	Gage
Minor					
201	Kink radius missing or improper		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9299108
4.4.2.13	Case, Ignition				NEXT HIGHER ASSEMBLY 9299111
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter, internal thread		0.40%	3.2	Gage
102	Minor diameter, internal thread		0.40%	3.2	Gage
103	Pitch diameter, external thread		0.40%	3.2	Gage
104	Major diameter, external thread		0.40%	3.2	Gage
105	Diameter (2 holes)		0.40%	3.2	Gage
106	Location of holes		0.40%	3.2	Gage
107	Inside diameter of cavity		0.40%	3.2	Gage
108	Outside diameter, max.		0.40%	3.2	Gage
109	Runout of outside thread with outside diameter and shoulder forward of aft thread		0.40%	3.2	Gage
110	Runout of inside thread with outside diameter and shoulder forward of aft thread		0.40%	3.2	Gage
111	Total length		0.40%	3.2	Gage
112	Diameter of inside cavity groove		0.40%	3.2	Gage
113	Diameter of aft tunnel cavity		0.40%	3.2	Gage
114	Length from aft face to shoulder		0.40%	3.2	Gage
115	Width of inside cavity groove		0.40%	3.2	Gage
116	Length from forward face to aft edge of inside cavity groove		0.40%	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.13	Case, Ignition				9299108
					NEXT HIGHER ASSEMBLY 9299111
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Depth of inside cavity		0.65%	3.2	Gage
202	Runout of aft tunnel cavity with outside diameter		0.65%	3.2	Gage
203	Diameter of case forward of aft thread		0.65%	3.2	Gage
204	Inside cavity groove radii missing or improper		0.65%	3.2	Gage
205	Runout of inside cavity diameter with outside diameter		0.65%	3.2	Gage
206	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.14	Basin (Type 1, 2, & 3)		1 OF 1		9299095
					NEXT HIGHER ASSEMBLY
					9299080
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Wall thickness		0.40%	3.2	Gage
103	Height		0.40%	3.2	Gage
Minor					
201	Radius missing or incomplete		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Basin (Type 4)	1 OF 1	9299100
			NEXT HIGHER ASSEMBLY: 9331155

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Wall thickness		0.40%	3.2	Gage
103	Height		0.40%	3.2	Gage
104	Outside diameter of raised hat		0.40%	3.2	Gage
Minor					
201	Presence of radii		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Nozzle, Jet	1 OF 1	9299089
			NEXT HIGHER ASSEMBLY 9299088

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter, thread		0.40%	3.2	Gage
102	Major diameter, thread		0.40%	3.2	Gage
103	True position of holes (8 holes)		0.40%	3.2	Gage
104	Runout of inner diameter with aft face and thread		0.40%	3.2	Gage
105	Inner diameter		0.40%	3.2	Gage
106	Diameter of holes		0.40%	3.2	Gage
107	Inside diameter of forward cavity		0.40%	3.2	Gage
108	Forward outside diameter		0.40%	3.2	Gage
109	Depth of forward cavity		0.40%	3.2	Gage
110	True position of forward outside diameter with inside diameter		0.40%	3.2	Gage
111	Overall length		0.40%	3.2	Gage
112	Length from aft face to seat		0.40%	3.2	Gage
Minor					
201	True position of forward cavity with inside diameter		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9299090
4.4.2.17	Disc				NEXT HIGHER ASSEMBLY 9299088
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40%	3.2	Gage
102	Diameter		0.40%	3.2	Gage
103	Diameter of boss		0.40%	3.2	Gage
104	Height of boss		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Foil, Damming	1 OF 1	9299093
			NEXT HIGHER ASSEMBLY 9299088

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Inner diameter		0.40%	3.2	Gage
103	Adhesive coating missing or incomplete		0.40%	3.2	Visual
104	Thickness of foil		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.19	Jet Nozzle Assembly	1 OF 1	9299088
			NEXT HIGHER ASSEMBLY: 9299080, 9331155

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Crimp inadequate or cracked		0.40%	3.2	Visual
102	Disc insecure or missing		0.40%	3.2	Visual
103	Cap missing, deformed or loose		0.40%	3.2	Visual
104	Foil missing, deformed or loose		0.40%	3.2	Visual
105	Lacquer not complete for 360°		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.20	Jet Housing Assembly (Type 1, 2 & 3)				9299080
					NEXT HIGHER ASSEMBLY 9299079
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight, propelling charge		100%	3.2	Gage
102	Weight, ejection charge		100%	3.2	Gage
103	Components improperly seated, loose, missing or deformed		0.40%	3.2	Visual/Manual
104	Flat disc missing or insecure		100%	3.2	Visual/Manual
105	Lacquer missing or improperly applied		0.40%	3.2	Visual
106	Torque improper		0.40%	3.2	Visual
107	Protective cap improperly sealed		0.40%	3.2	Visual
108	Notches - proper number for given charge		100%	3.2	Visual
109	Basin flush or below flush with jet body		0.65%	3.2	Visual
110	Charge weight assessment	30(a)		3.2	4.5.4
Minor					
201	Excessive adhesive (jet screw threads)		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

(a) Charge weight assessment shall be performed for each new propellant and/or ejection charge black powder lot .

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	Cap, Protective	1 OF 1	9299097
			NEXT HIGHER ASSEMBLY
			9299080/9331155

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter of cavity		0.40%	3.2	Gage
102	Depth of cavity		0.40%	3.2	Gage
103	Cavity radius missing or improper		0.40%	3.2	Visual
104	Cavity radius		0.40%	3.2	Gage
105	Forward outside diameter		0.40%	3.2	Gage
Minor					
201	Presence of finger grips		0.65%	3.2	Visual
202	Presence of flash inside cavity		0.65%	3.2	Visual
203	Height of cap		0.65%	3.2	Gage
204	Rear outside diameter (hub)		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.22	Cartridge 22MM, Sub-Caliber Practice M744, M745, M746, M747		1 OF 1	9299079
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of safety ball		0.40%	3.2	Gage
102	Forward outside diameter, max.		0.40%	3.2	Chamber gage
103	Jet housing assembly torque		0.40%	3.2	Gage
104	Head torque		0.40%	3.2	Gage
105	Overall length, max.		0.40%	3.2	Gage
106	True position of jet housing assembly barrel outside diameter to case cartridge assembly outside diameter		0.40%	3.2	Gage
107	Head not painted or improperly painted		0.40%	3.2	Visual
108	Marking missing or incomplete		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.23	Tray Assembly for Cartridge Subcaliber, 22MM Practice		1 OF 1	9322198
				NEXT HIGHER ASSEMBLY
				9322202

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cover and body do not mate		0.40%	3.2	Visual
102	Assembly damaged		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Body, Jet	1 OF 2	9299084
			NEXT HIGHER ASSEMBLY
			9299080/9331155

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		0.40%	3.2	Gage
102	Major diameter of forward thread		0.40%	3.2	Gage
103	Pitch diameter of forward thread		0.40%	3.2	Gage
104	Minor diameter of aft thread		0.40%	3.2	Gage
105	Pitch diameter of aft thread		0.40%	3.2	Gage
106	Runout of aft thread with dia. forward of shank and aft face		0.40%	3.2	Gage
107	Runout of forward thread with dia. forward of shank and aft thread relief surface		0.40%	3.2	Gage
108	Outside diameter of shank		0.40%	3.2	Gage
109	Outside diameter forward of shank		0.40%	3.2	Gage
110	Outside diameter aft of shank		0.40%	3.2	Gage
111	Length from forward face to first step		0.40%	3.2	Gage
112	Length from forward face to large diameter shank step		0.40%	3.2	Gage
113	Length from aft face to large diameter shank step		0.40%	3.2	Gage
114	Diameter of tunnel		0.40%	3.2	Gage
115	Length of forward cavity		0.40%	3.2	Gage
116	Length of aft cavity		0.40%	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Body, Jet	2 OF 2	9299084
			NEXT HIGHER ASSEMBLY 9299080/9331155

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Length of forward thread relief		0.65%	3.2	Gage
202	Presence of aft face angled notch		0.65%	3.2	Visual
203	Diameter of forward cavity		0.65%	3.2	Gage
204	Forward or aft thread chamfers missing		0.65%	3.2	Visual
205	Diameter of aft cavity		0.65%	3.2	Gage
206	Diameter of aft cavity groove		0.65%	3.2	Gage
207	Length of aft cavity groove		0.65%	3.2	Gage
208	Length from aft face to forward edge of aft cavity groove		0.65%	3.2	Gage
209	Forward thread relief diameter		0.65%	3.2	Gage
210	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER
4.4.2.25	Charge, Ejection				9299098
					NEXT HIGHER ASSEMBLY
					9331155/9299080
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Composition improper	2 grams		3.2	Gage
102	Granulation size improper	2 grams		3.2	Gage
Minor					
201	Evidence of poor workmanship	2 grams		3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9299091
4.4.2.26	Cap, Percussion				NEXT HIGHER ASSEMBLY 9299088
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter at flange		0.40%	3.2	Gage
102	Outside diameter of battery cup		0.40%	3.2	Gage
103	Overall length of cap		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9299096
4.4.2.27	Disc, Flat				NEXT HIGHER ASSEMBLY 9299080/9331155
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of disc		0.40%	3.2	Gage
102	Thickness of disc		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.28	Cap, Percussion				9299103
					NEXT HIGHER ASSEMBLY 9299101
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter at flange		0.40%	3.2	Gage
102	Overall length		0.40%	3.2	Gage
103	Diameter of primer cup		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9299104
4.4.2.29	Disc, Seal				NEXT HIGHER ASSEMBLY 9299112
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of disc		0.40%	3.2	Gage
102	Thickness of disc		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.30	Disc	1 OF 1	9299107
			NEXT HIGHER ASSEMBLY 9299111

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of disc		0.40%	3.2	Gage
102	Thickness of disc		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.31	Smoke Charge Assembly		1 OF 1	9299111
				NEXT HIGHER ASSEMBLY 9299112

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		0.40%	3.2	Gage
102	Ignition case improperly torqued to case assembly		0.40%	3.2	Gage
103	Improper assembly of components		0.40%	3.2	Visual
104	Smoke charge missing		0.40%	3.2	Radiographic
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9299113
4.4.2.32	Charge, Smoke				NEXT HIGHER ASSEMBLY 9299111
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Mixture improper	20 grams		3.2	Visual
Minor 201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.33	Jet Housing Assembly (Type 4)				9331155
					NEXT HIGHER ASSEMBLY
					9299079
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight, propelling charge		100%	3.2	Gage
102	Weight, ejection charge		100%	3.2	Gage
103	Components improperly seated, loose, missing or deformed		0.40%	3.2	Visual/Manual
104	Flat disc missing or insecure		100%	3.2	Visual/Manual
105	Lacquer missing or improperly applied		0.40%	3.2	Visual
106	Torque improper		0.40%	3.2	Visual
107	Protective cap improperly sealed		0.40%	3.2	Visual
108	Notches - proper number		100%	3.2	Visual
109	Basin flush or below flush with jet body		0.65%	3.2	Visual
110	Charge weight assessment	30(a)		3.2	4.5.4
Minor					
201	Excessive adhesive (jet screw threads)		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES: (a) Charge weight assessment shall be performed for each new propellant and/or ejection charge black powder lot.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.34	Packing and Marking for Box, Wood, Wirebound, for Cartridge, Subcaliber 22MM, Practice	1 OF 1	9322202
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged		0.40%	3.2	Visual
Minor					
201	Marking missing, illegible, or incomplete		0.65%	3.2	Visual
202	Barrier bag torn or improperly sealed		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112 (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.35	Proving Ground Test Cartridge 22MM, Sub-Caliber Practice M744, M745, M746, M747		1 OF 1	9299079
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	for ea. part #:			
Special					
a	Short round	25	(a)	3.3	4.5.2
Major					
101	Misfire	25	(b)	3.3	4.5.2
102	Cartridge fails to function audibly	25	(b)	3.3	4.5.2
103	Smoke charge failure	25	(b)	3.3	4.5.2
104	Premature burst	25	(b)	3.3	4.5.2
105	Duds	25	(c)	3.3	4.5.2
Minor	None defined				

NOTES: (a) One or more defects shall be cause for lot rejection. Wind speed shall be less than 10 MPH at time of firing. (b) Two or more defects shall be cause for lot rejection. (c) Five or more duds shall be cause for the lot to be placed in referred status. ARRADCOM shall be contacted to the attention of DRDAR-QAR-M when this occurs.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, sub-assemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table 1.

4.5.2 Proving ground test

4.5.2.1 Functioning test. The test shall be performed at a Government Proving Ground in accordance with the applicable USATECOM Acceptance Test Procedure for standard 81MM mortar testing. Use six sabots in succession. Sabots will not require conditioning. The cartridges shall be conditioned a minimum of six (6) hours at 70°F and test fired within two (2) minutes after removal from the conditioning chamber. The individual velocities, deflections, ranges, sound and smoke functioning shall be recorded on the firing record. Temperature of the day, wind and weather conditions shall also be recorded. Production lot testing shall utilize twenty five (25) rounds at each charge level (1 to 4) with a firing elevation of 800 mils.

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4.5.2.2 Transportation-vibration. The cartridges in the packaged condition shall be subjected to the required vibration test, (see para. 3.3.5) conditioned at range ambient temperature and test fired per the procedure of 4.5.2.1.

4.5.3 Check test for deterioration of primers. If the total time between original acceptance of any primer lot and the assembly of that lot into the cartridges exceeds two years, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass the functioning and output tests specified in MIL-P-48077 immediately before the primer lot is assembled into the cartridges. This test shall be performed by the contractor on primers selected by the Government Inspector at the facility assembling the primers into the cartridges. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or the contractor who assembled the primers into the cartridges. The test shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer was under the jurisdiction of either contractor.

4.5.4 Propellant charge ejection charge weight assessment. (To be conducted at a US Government Proving Ground) Each black powder lot of propellant and ejection charge shall be assessed. Ten each (thirty total for each of the following part numbers 9299080-1, 9299080-2, 9299080-3, 9331155) ignition cartridges shall be loaded at charge weight noted below. The ignition cartridges shall be assembled to 22MM subcaliber production rounds and 81MM sabots. Rounds shall achieve a total weight of 9.31 pounds plus of minus 0.25 pounds. The rounds shall be conditioned at 70°F (+2.5°F) for 6 hours prior to firing and shall be maintained at this temperature until fired. The rounds shall be fired from an 81MM ballistic tube at an elevation of 800 mils. Calibration rounds shall be fired alternately for velocity correction. The mean corrected velocities of the four groups shall be used to establish the charge weight velocity relationship and the charge weight assessment shall be determined to the nearest grain.

Charge Prop Weights

Jet housing assembly	Weights		
9299080-1	0.62 grams	0.70 grams	0.78 grams
9299080-2	0.82 grams	0.90 grams	0.98 grams
9299080-3	1.12 grams	1.20 grams	1.28 grams
9331155	1.62 grams	1.70 grams	1.78 grams

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4.5.5 Wing embrittlement. - The wing shall be bent until it no longer returns to its original position. When the spring no longer returns to its original profile, it shall be considered distorted permanently. Any wing that breaks prior to this point shall be classed defective.

5. PACKAGING

5.1 Packaging requirements. Packaging shall be as specified on dwg. 9322198, 9322199, and 9322201.

5.2 Packing and marking. Packing and marking shall be as specified on dwg. 9322202.

6. NOTES

6.1 Intended use. The cartridges covered by this specification are intended for use on the 81MM Mortar Training Device.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover NJ 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Data requirements. - Deliverable data required by this specification are in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.5.5	Quality Inspection Defect Report	DI-R-1721

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(Copies of data items description required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

CUSTODIAN:
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PREPARING ACTIVITY:
Army-AR

Project Number: 1305-A848

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